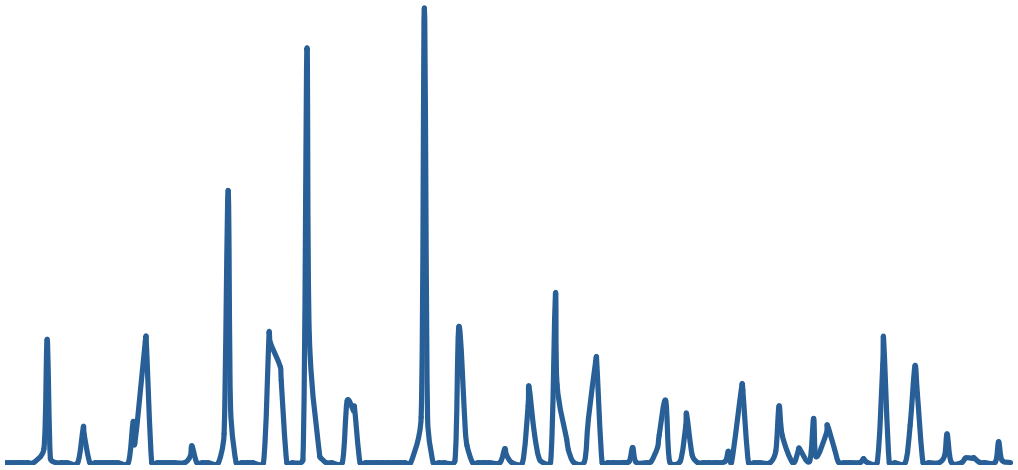
Research task: identify crystal systems from XRD patterns

XRD pattern



→

Crystal-system identification

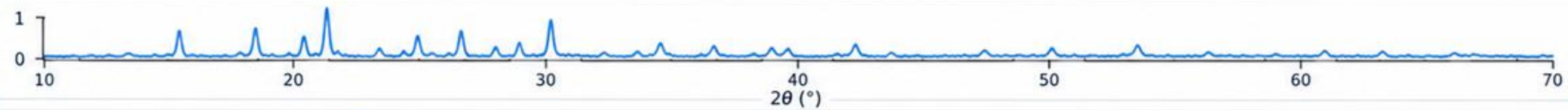
One of seven crystal systems

Cubic · Tetragonal · Orthorhombic · Hexagonal
Trigonal · Monoclinic · Triclinic

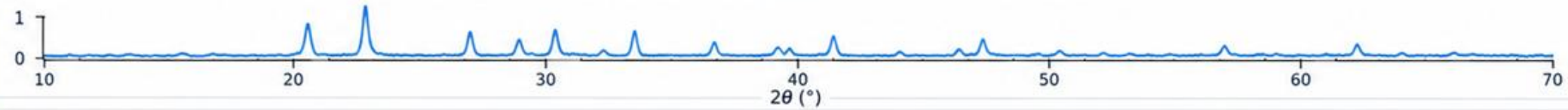
Representative PXRD patterns for the seven crystal systems

Note: These are representative examples, not fixed XRD fingerprints. Crystal-system information comes mainly from overall peak positions, relationships among peaks, and symmetry.

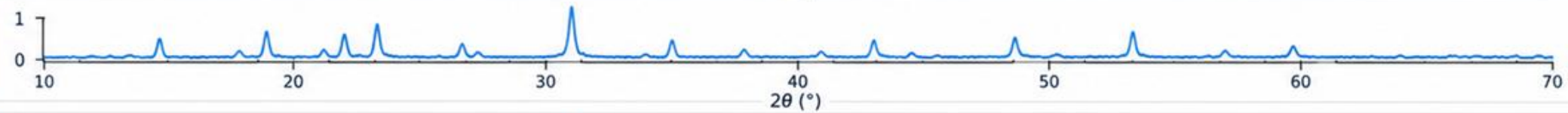
Triclinic



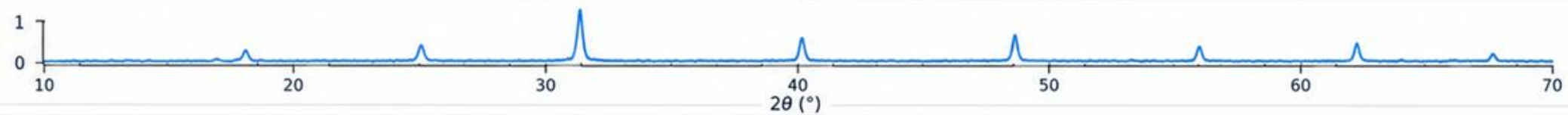
Monoclinic



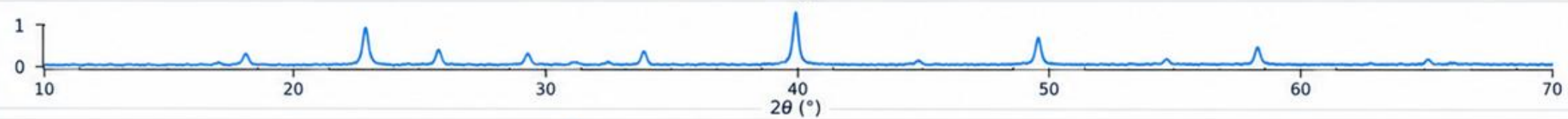
Orthorhombic



Tetragonal



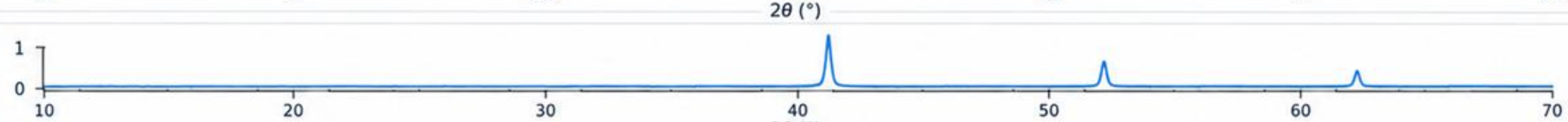
Trigonal / rhombohedral



Hexagonal

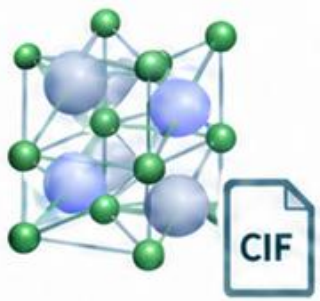


Cubic



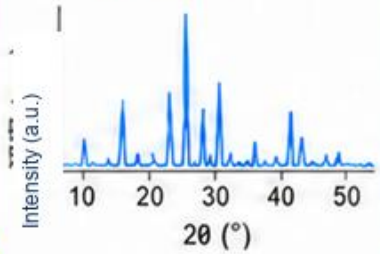
Overall project workflow

1 Crystal structure / CIF data



- Input: crystal structure (CIF)
- Split by crystal structure into train / validation / test
- Target task: seven-class classification
- One crystal-system label per structure

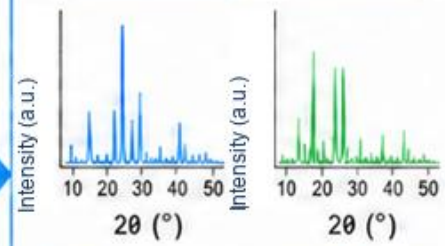
2 Online physical simulation of PXRD



- Generate patterns online during training, not from a fixed offline library.
- Five measurement perturbations:
  - Peak shift
  - Peak intensity
  - Background
  - Noise
  - Preferred orientation
- Expand simulated-domain coverage.

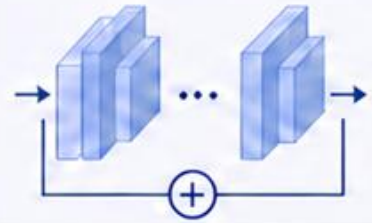
3 Same-structure dual-view construction

View 1 | View 2



- One structure under different measurement conditions produces two patterns.
- Both patterns share one crystal-system label.
- Pair them and ask the model for similar predictions.

4 1D ResNet training



- **CE classification loss**
  Predict one of seven crystal systems.
- **JS consistency constraint**
  Two patterns from one structure should produce similar predictions.

Core idea: use the same-source relationship, not only class labels.

Simulated-domain evaluation

- Evaluate standard simulated tests and split generalization.
- Focus on robustness under each perturbation.

Peak shift | Peak intensity | Noise
Background | Preferred orientatior | Texture

- Compare method performance.

Real experimental-domain validation

- Zero-shot external validation
- Few-shot real-domain adaptation
- Simulation-to-real transferability
- Compare with conventional methods

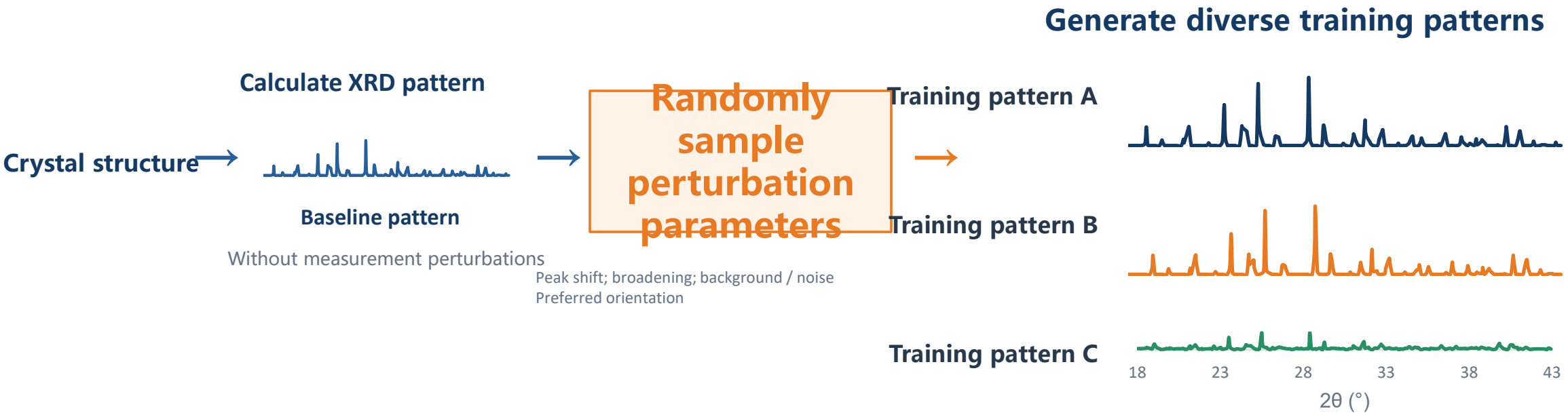
CNRS dataset | RRUFF dataset

Overview: This project studies how online physical simulation and same-structure consistency supervision can improve robustness and generalization in seven-system PXRD classification under simulated perturbations and real experimental conditions.

Workflow: Structure -> simulation -> paired views -> training -> simulated evaluation / real validation

Prior work: randomize measurement perturbations to generate diverse training patterns

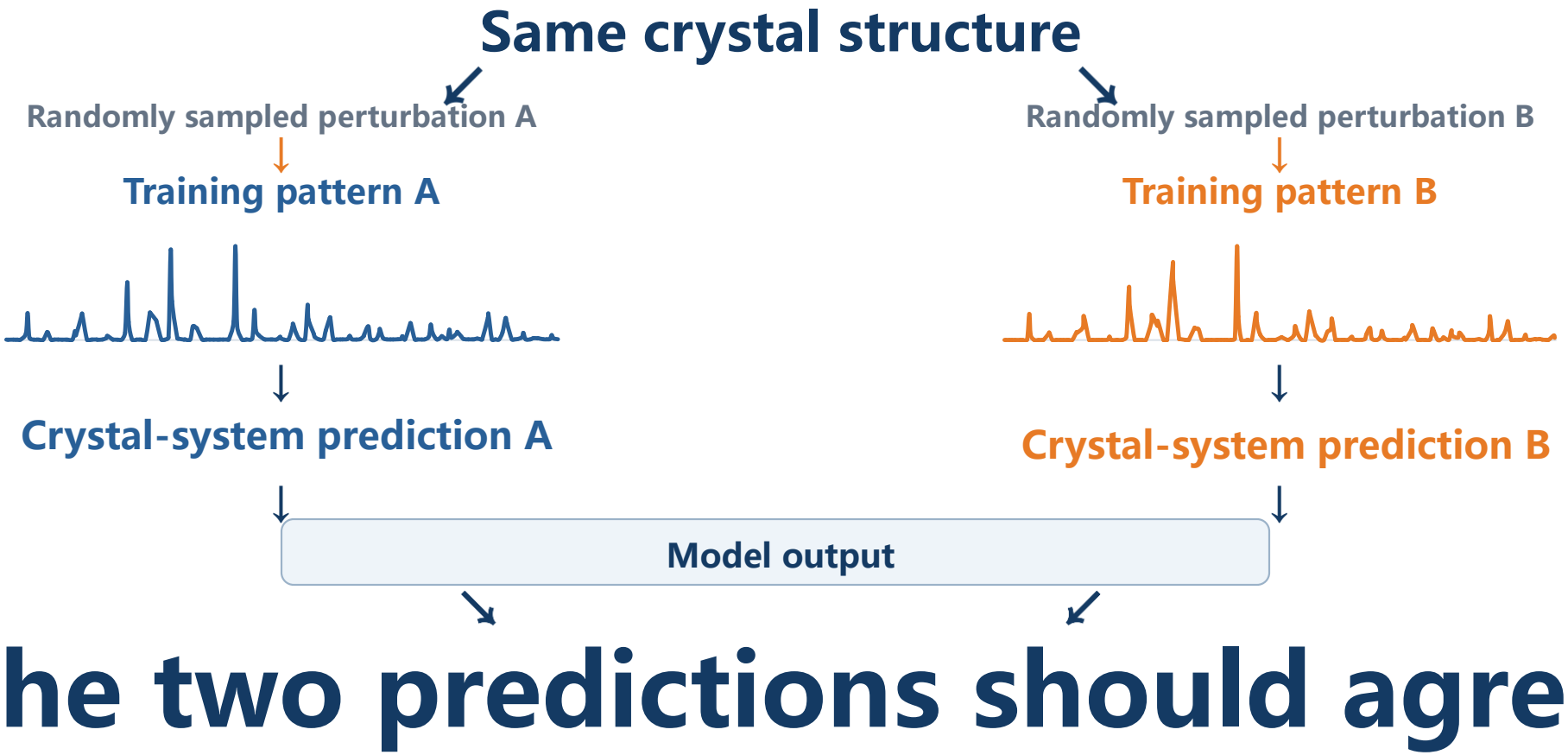
Prior work samples parameters for peak shift, broadening, background / noise, and preferred orientation to generate diverse training patterns



But patterns generated from the same source are usually still treated as independent training samples.

Selected studies: Oviedo et al. (2019); Szymanski et al. (2021); Lee et al. (2023); Schopmans et al. (2023)

Our improvement: the predictions for the same structure should agree under different perturbations



Core idea: encode the physical relation “measurement conditions can change, structure does not” in model training.

Data and experimental setup: 14,060 crystal structures and five measurement perturbations

14,060

Crystal structures from Materials Project

Train / validation / test

9,842 / 2,109 / 2,109

Split by crystal structure

Different patterns from the same crystal always stay in the same split

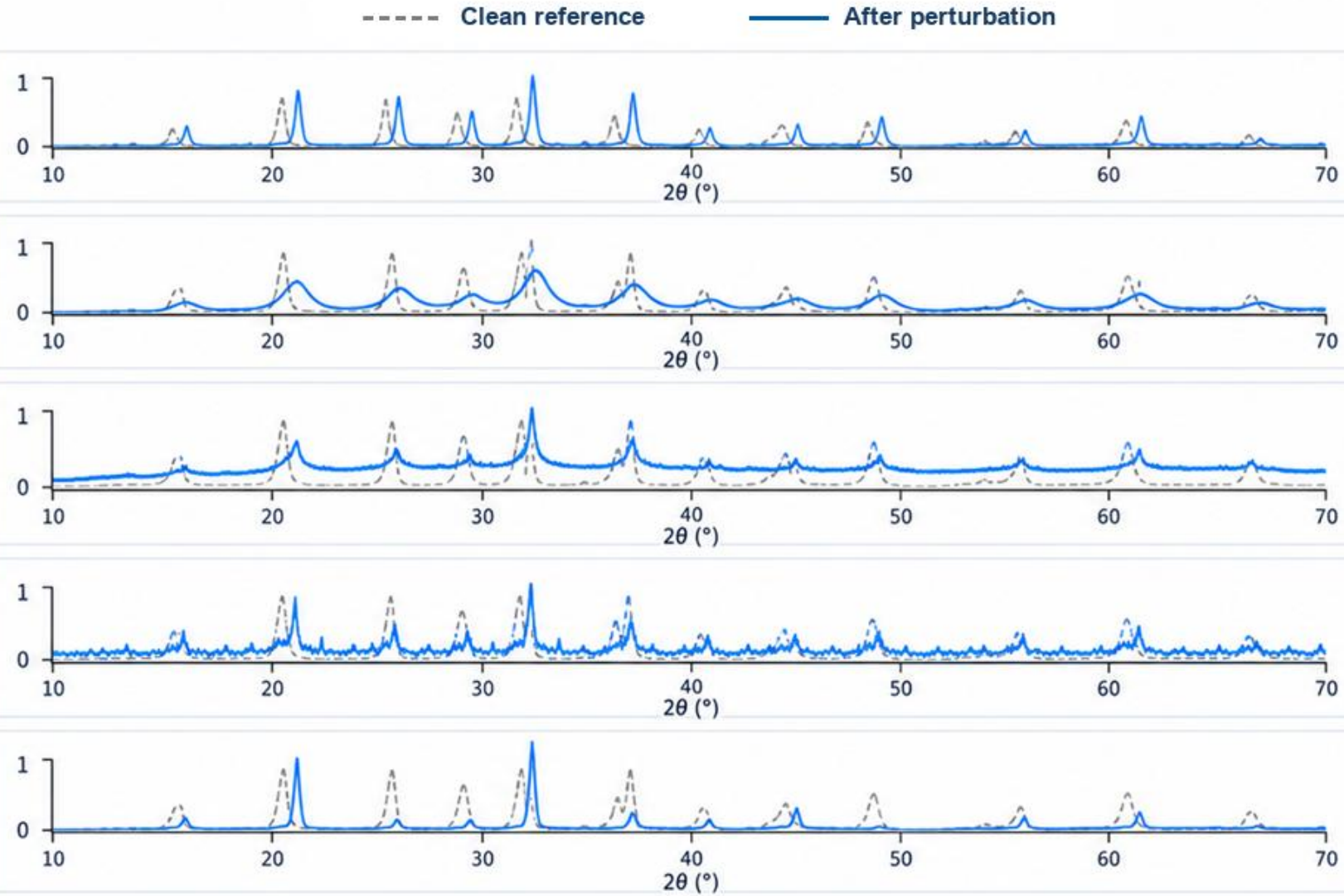
Patterns generated online during training

Same crystal → sample a new set of measurement perturbations each time → generate a different training pattern

Perturbation type	Effect on the pattern
Peak shift	All peaks move horizontally
Peak broadening	Peaks widen; neighboring peaks overlap more easily
Background change	The baseline or background shape changes
Counting noise	Random fluctuations appear in the pattern
Preferred orientation	Relative diffraction-peak intensities change

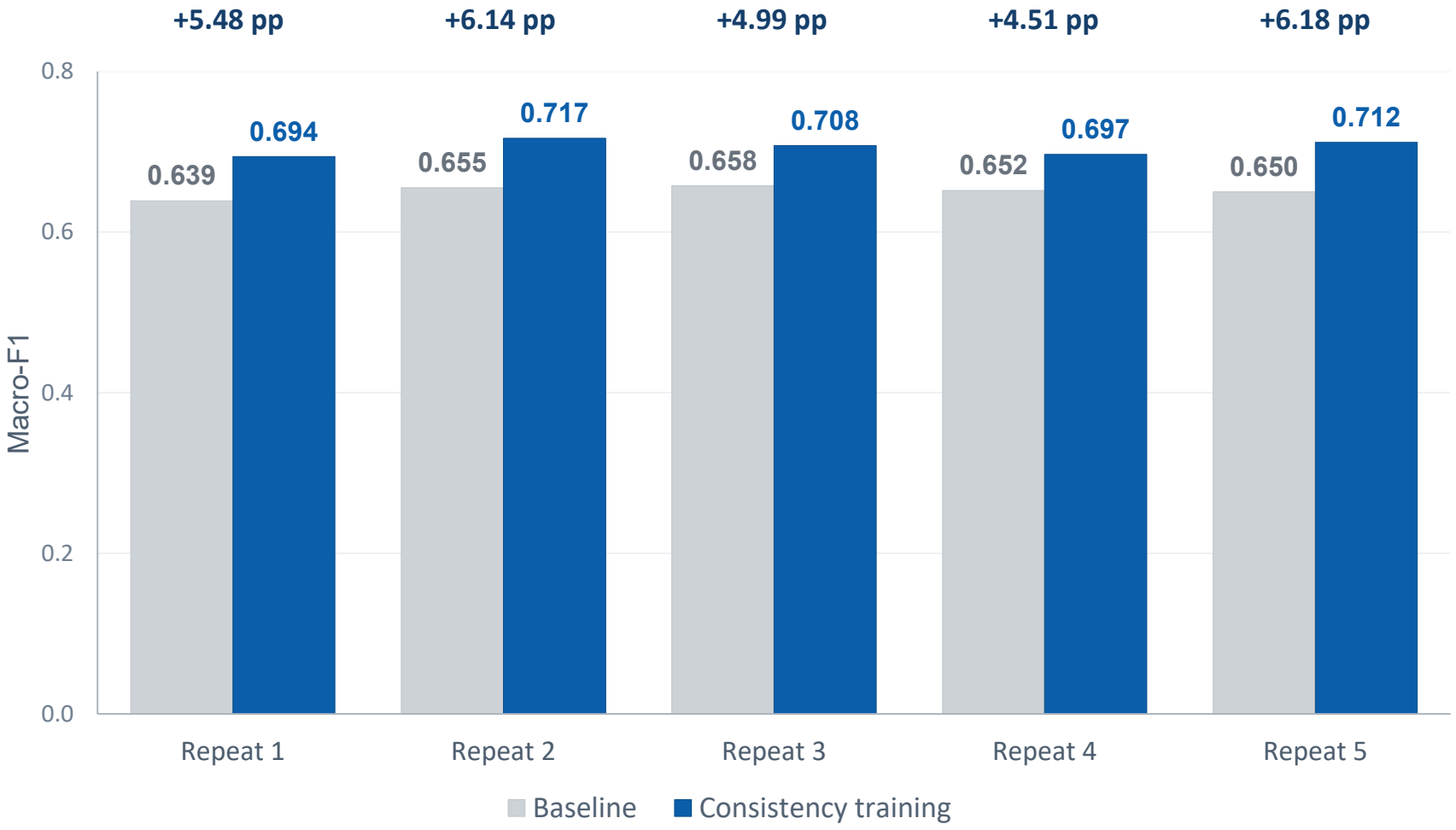
PXRD changes for the same crystal under five perturbations

The same parent structure is held fixed while measurement / sample conditions change. Gray dashed: clean reference; blue solid: after perturbation.



1. Simulated results: improvements in all five training runs

Test set: 2,109 held-out MP structures; test patterns use fixed perturbations outside the training range.



5 / 5 ↑

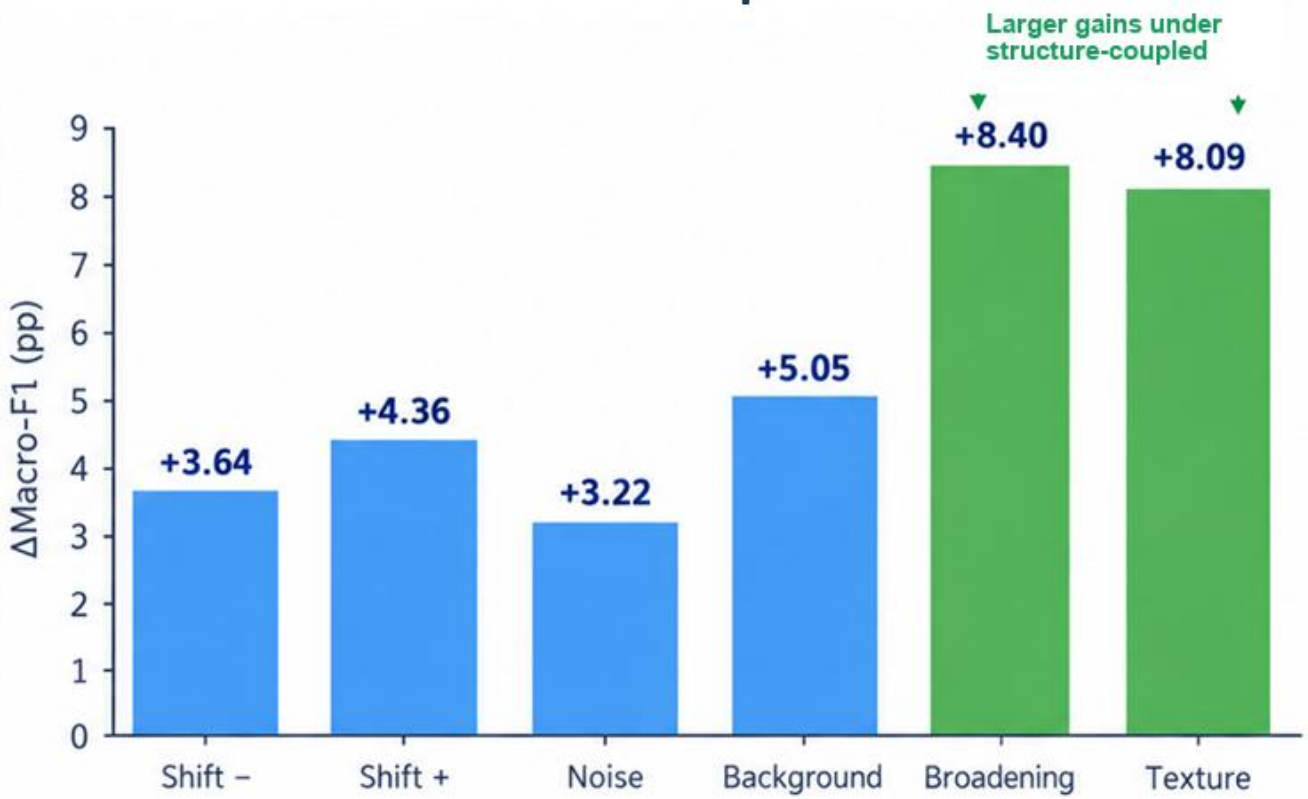
Average +5.46 pp

With the same data and training settings, all five repeated runs improved.

Consistency gains are larger for structure-coupled perturbations

Broadening and texture depend on peak spacing, overlap, and hkl reflection-intensity patterns.

ΔMacro-F1 across six OOD perturbations



Takeaway:

Same-parent consistency helps most when measurement perturbations are more strongly coupled to the underlying crystal

Weak structure coupling: shift / noise / background

Mainly affect measurement coordinates, random noise, or baseline shape.

- Shift − / +: all peaks move along 2θ; dependence on specific peak shape is relatively weak.
- Noise: adds random intensity fluctuations.
- Background: raises or distorts the baseline.

Strong structure coupling: broadening / texture

Effect size depends on the original peak pattern.

- Broadening: at the same width, peaks overlap more in structures with small peak spacing / dense peaks.
- Texture: preferred orientation selectively changes hkl reflection intensities; the effect depends on the original reflection combination.

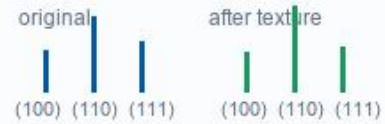
Why are these perturbations structure-coupled?

A Broadening



Same broadening width; overlap depends on peak spacing.

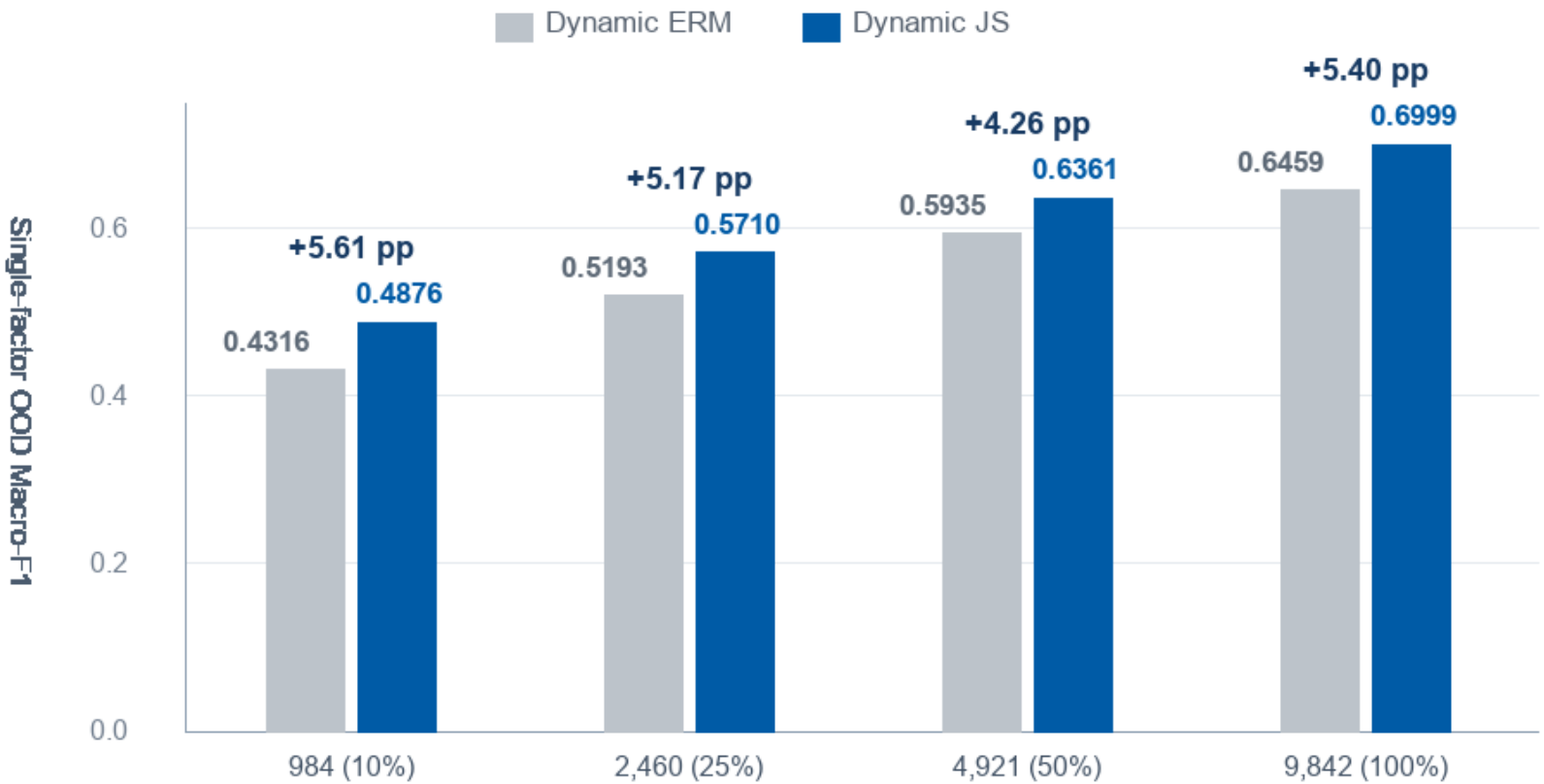
B Texture



Different hkl patterns produce different intensity changes.

2. Data efficiency: JS improves Macro-F1 at matched budgets

Simulated single-factor OOD Macro-F1 | 10%, 25%, 50%, and 100% of training parents



Matched-budget gains are shown; parent savings are not quantified.

① JS exceeds ERM at every matched budget

5 / 5 paired seeds positive at each budget

Paired gains: +5.61, +5.17, +4.26, +5.40 pp

② Equal training exposure

40 / 40 models completed

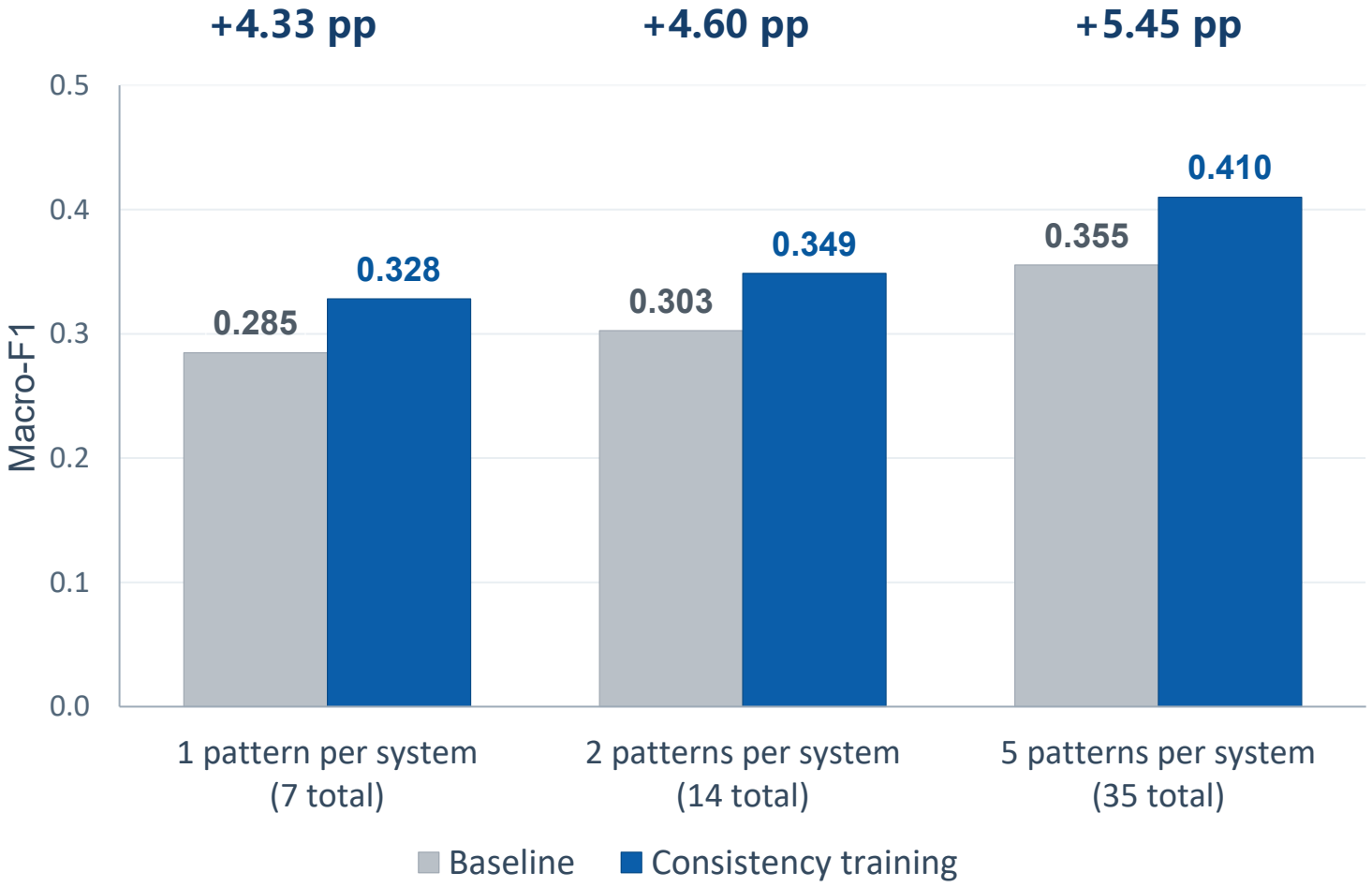
49,280 optimizer steps per model

Supplementary evaluation

Five paired seeds per budget. The supplement reuses the previously evaluated simulated Test (post-main).

3. RRUFF-301: few-shot fine-tuning with a small number of real patterns

RRUFF-301: 301 experimental patterns | Fine-tune with 1 / 2 / 5 patterns per crystal system



① All three real-label budgets improved

Average gain: about +4.8 pp

② With fewer real patterns, this method reached similar or better performance

This method: 14 patterns ≈ baseline: 35 patterns

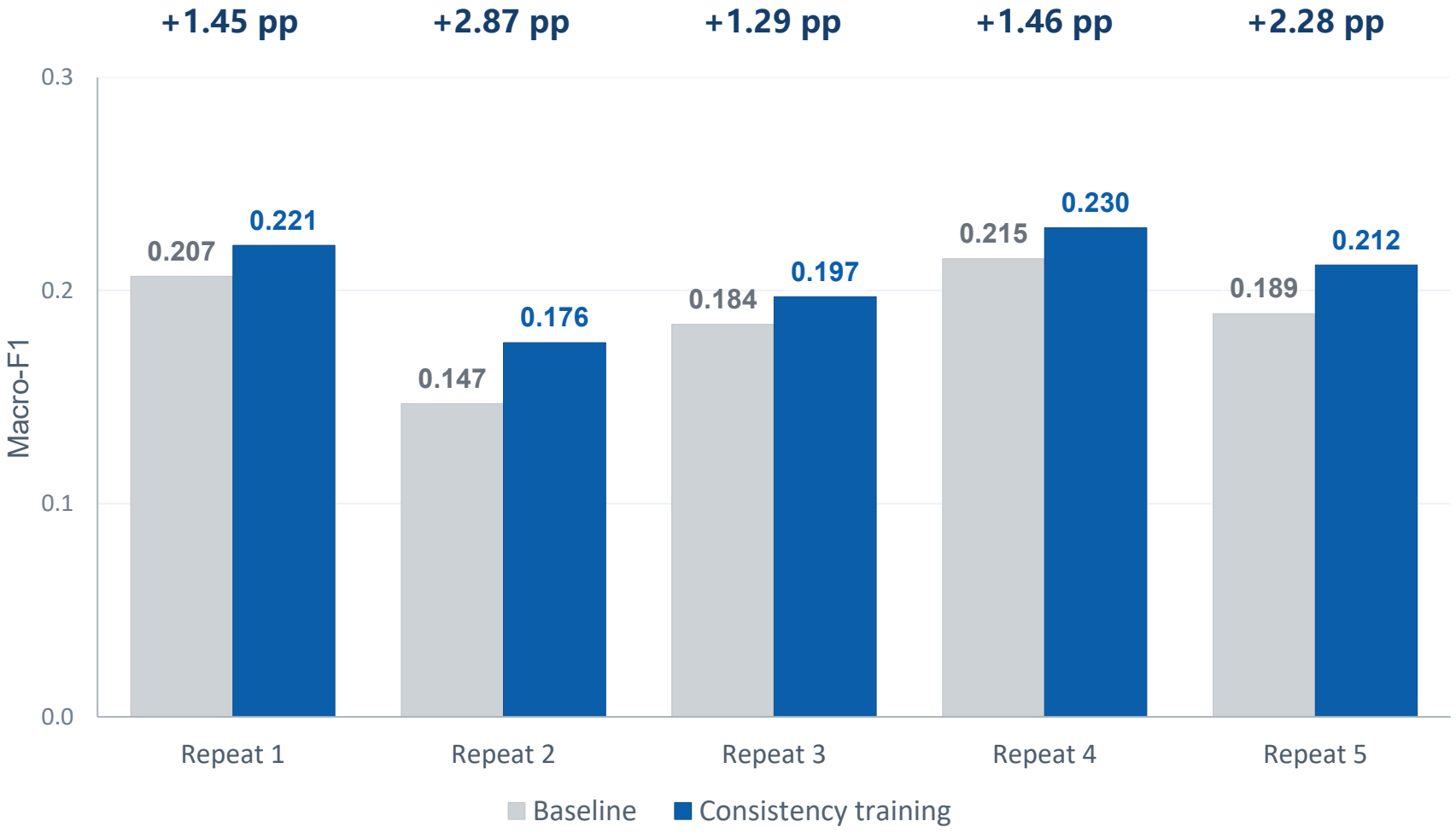
0.349 ≈ 0.356

This method: 7 patterns > baseline: 14 patterns

0.328 > 0.303

4. Independent CNRS-318 zero-shot results: all five runs improved

CNRS-318: an independent set of experimental patterns | Test directly without adapting the model on CNRS data



5 / 5 ↑

All five repeated runs improved

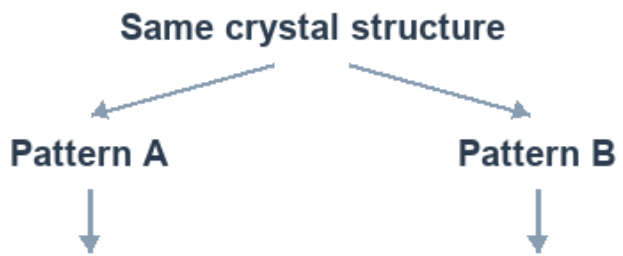
Average +1.87 pp

The model was tested directly on CNRS data without adaptation and still showed a gain.

Next stage: extend the same physical-constraint idea from classification to quantitative inversion

Current: qualitative classification

For different measurements of the same structure, the model predictions should agree.



The two predictions should agree

Physical relation: measurement conditions can change; crystal structure remains the same.

Qualitative classification

Quantitative inversion

Next step: quantitative inversion

If the inverted parameters are reasonable, they should regenerate and explain the input pattern.

Input XRD pattern

↓

Infer physical parameters

a, c, zero-point shift, FWHM

↓

Generate a calculated pattern from the parameters

↓

Generated pattern approximately matches input pattern

Physical relation: when parameters are reasonable, the generated pattern should match the experimental pattern.

Unified idea: encode known physical relations as constraints on model training and result interpretation.